1. Technical Field:
This falls within the marine equipment manufacturing sector and serves the marine shellfish aquaculture and processing industries.

2. Background:
Marine shellfish farming and processing generate large quantities of shell waste; traditional disposal methods (landfilling or reclamation) pose environmental pollution issues.
Shells contain calcium carbonate, which can be converted into valuable calcium powder through a series of processes for use in various industries.

3. Summary of the Invention:
A new type of equipment has been designed to simultaneously wash and spin-dry shells within a single unit, thereby improving processing efficiency.
The structure of the equipment comprises a feed inlet, a discharge outlet, a cylindrical outer casing, an internal rotating drum and a support frame, and is equipped with variable-frequency speed control.
The rotating drum is constructed from 304 stainless steel and features specific screen apertures and rubber (or nylon) columns to enhance friction and agitation for shell removal.
The high-pressure rinsing system is fitted with multi-stage pumps, nozzles and pipework; wastewater is recycled to minimise environmental pollution.

4. Specific Example:
The equipment is designed to process 12 tonnes of shell waste per hour, with a daily processing capacity of approximately 300 tonnes.
The motor has a power rating of 5 kW, with a rinsing water pressure of 3 kg and a flow rate of 30 cubic metres per hour.
An oxidising agent solution and caustic soda flakes are used as rinsing agents.
Through the operation of the equipment, the shells are effectively cleaned, and the moisture content is controlled to within 3 per cent after centrifugation.

The composition and functions of the equipment are described, including the feed inlet, cylindrical body, rotating drum, rubber columns, screen apertures and discharge outlet. Emphasis is placed on the equipment’s vibration-damping design, balancing and commissioning, the water recycling system, and automated operation controlled by a CNC cabinet.
This equipment is designed to address the environmental issues associated with shell waste whilst creating economic value through resource recovery.